



Modelling Financial Markets

Chapter 4: Portfolio Optimisation



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Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ DSR: Deflated Sharpe Ratio
- ▣ ERC: Equal Risk Contribution portfolio
- ▣ FDR: False Discovery Rate
- ▣ FWER: Family-Wise Error Rate
- ▣ GLS: Generalised Least Squares
- ▣ GMV: Global Minimum-Variance portfolio
- ▣ GMV-LO: Global Minimum-Variance portfolio, Long-Only
- ▣ GMV-LW: Global Minimum-Variance portfolio with Ledoit-Wolf covariance
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HRP: Hierarchical Risk Parity
- ▣ HYG: iShares iBoxx High Yield Corporate Bond ETF
- ▣ IEF: iShares 7–10 Year Treasury Bond ETF
- ▣ LO: Long-Only, no short sales
- ▣ LQD: iShares iBoxx Investment Grade Corporate Bond ETF
- ▣ ML: Machine Learning
- ▣ MV: Mean–Variance tangency portfolio estimated from sample moments
- ▣ MV-LO: Mean–Variance tangency portfolio, Long-Only
- ▣ NYSE: New York Stock Exchange
- ▣ OLS: Ordinary Least Squares
- ▣ OMV: Österreichische Mineralölverwaltung (Austrian Mineral Oil Administration (OMV group))
- ▣ OOS: Out-Of-Sample

Acronyms used in this material (2/2)

- POET: Principal Orthogonal compleMent Thresholding
- US: United States

Chapter Route and Outcomes

The question

- A **portfolio** is a weight vector $w = (w_1, \dots, w_N)^\top$ with $w^\top \mathbf{1} = 1$; optimisation chooses w to maximise a criterion under constraints
- The optimum is easy to write with known means and covariances, hard to estimate: the optimiser loads on the largest estimation errors
 - ▶ 16 assets: $16 + 136 = 152$ parameters, estimated from 60 monthly observations of a 16-dimensional return vector
- Can any estimated rule beat equal weights out of sample, after costs?

Route

- Part I: mean–variance theory, estimation risk, covariance shrinkage, Black–Litterman
- Part II: risk parity and HRP, out-of-sample tests, BVB portfolios, a case study, one pass of the AI-assisted research loop
- Technical detail sits in the [Appendix](#); textbook: [Franke, Härdle & Hafner \(2019\)](#), Part I

Data, Assets and the 1/N Benchmark

Universe	Assets	Returns	Currency
Sectors	9 US sector ETFs: XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, XLY	1999/01 – 2026/07	USD
Multi-asset	SPY, EFA, EEM (equities); TLT, IEF (Treasuries); LQD, HYG (credit); GLD (gold)	2007/05 – 2026/07	USD
Combined	9 sectors + the 7 multi-asset ETFs other than SPY (16 ETFs)	2007/05 – 2026/07	USD
Romania	8 BVB blue chips: TLV, SNP, BRD, TGN, SNG, SNN, EL, TEL; BET, BET-TR	2014/10 – 2026/08	RON

- Monthly simple returns from month-end prices on common trading days; ETFs and stocks with dividends reinvested; BET is a price index, BET-TR reinvests dividends
- Excess returns over the one-month Treasury bill ([Kenneth French Data Library](#)); BVB: $R_f = 0$
- Sectors: XLB materials, XLE energy, XLF financials, XLI industrials, XLK technology, XLP staples, XLU utilities, XLV health care, XLY discretionary
- Multi-asset: SPY (S&P 500), EFA (developed-market equities outside the US and Canada), EEM (emerging-market equities), TLT (US Treasuries, 20+ years), IEF (US Treasuries, 7–10 years), LQD (investment-grade corporate bonds), HYG (high-yield corporate bonds), GLD (gold)
- BVB: TLV Banca Transilvania, SNP OMV Petrom, BRD BRD–Groupe Société Générale, TGN Transgaz, SNG Romgaz, SNN Nuclearelectrica, EL Electrica, TEL Transelectrica
- 1/N: equal weights $w_i = 1/N$, rebalanced every month; no estimation at all, the benchmark of every comparison

Notation

- Simple return $R_{i,t} = P_{i,t}/P_{i,t-1} - 1$; excess return $R_{i,t}^e = R_{i,t} - R_{f,t}$; portfolio mean $w^\top \mu$, variance $w^\top \Sigma w$
- **Sharpe ratio** ([Sharpe \(1966\)](#)): $SR_p = \mathbb{E}[R_p^e]/\sigma(R_p^e)$; monthly figures annualised with $\times 12$ (mean) and $\times \sqrt{12}$ (volatility, Sharpe)
- **Maximum drawdown**: $\min_t(W_t/\max_{s \leq t} W_s - 1)$, the largest fall of wealth W_t from a previous peak
- **Out of sample** (OOS): the weights held in month t use only data up to month $t - 1$
- Rules of this chapter: MV (tangency from sample moments), GMV (global minimum variance), their long-only versions MV-LO and GMV-LO, GMV-LW (GMV with shrinkage), ERC, HRP and 1/N

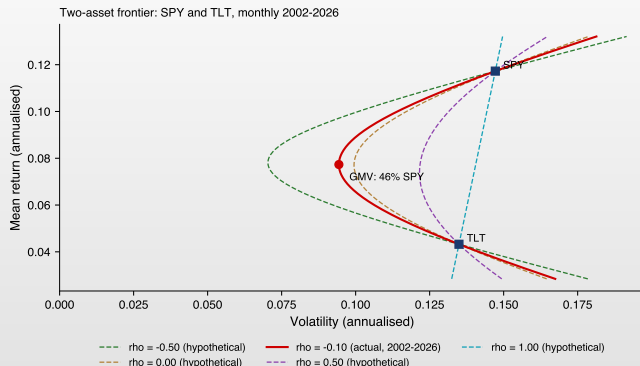
The Markowitz Problem

- ▣ **Markowitz (1952)**: investors care about the mean and the variance of portfolio returns
- ▣ **Minimum-variance frontier**: minimum variance for every target mean m ; its upper branch, $m \geq \mu_{\text{GMV}}$ (the mean of the GMV portfolio), is the **efficient frontier**

$$\min_w w^\top \Sigma w \quad \text{s.t.} \quad w^\top \mu = m, \quad w^\top \mathbf{1} = 1$$

- ▣ Utility form, risk aversion $\gamma > 0$: $\max_w w^\top \mu - \frac{\gamma}{2} w^\top \Sigma w$ s.t. $w^\top \mathbf{1} = 1$; as γ varies it traces exactly the efficient branch
- ▣ Weights may be negative: a **short sale** (selling a borrowed asset)

Two Assets: How Correlation Diversifies



- Two assets, correlation ρ :

$$\sigma_p^2 = w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\rho\sigma_1\sigma_2$$
- Minimised at $w^* = \frac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$
- SPY and TLT, monthly 2002–2026:
 $\rho = -0.10$, $w^* = 46\%$ SPY, volatility 9.4% against 14.7% and 13.5%
- The lower ρ , the more the curve bends to the left; at $\rho = 1$ there is no diversification
- Time-varying correlation: [Chapter 6](#)

GMV and Tangency Portfolios for N Assets

- GMV: $\min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w}$ s.t. $\mathbf{w}^\top \mathbf{1} = 1$; Lagrangian $\mathcal{L} = \mathbf{w}^\top \Sigma \mathbf{w} - \lambda(\mathbf{w}^\top \mathbf{1} - 1)$, Σ positive definite
 - First-order condition $2\Sigma \mathbf{w} = \lambda \mathbf{1}$, so $\mathbf{w} = \frac{\lambda}{2} \Sigma^{-1} \mathbf{1}$; the constraint gives $\frac{\lambda}{2} = 1/A$, $A = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$
- Tangency: maximise $\mathbf{w}^\top \boldsymbol{\mu} / \sqrt{\mathbf{w}^\top \Sigma \mathbf{w}}$, $\boldsymbol{\mu}$ the expected **excess** returns; the condition $\boldsymbol{\mu} \propto \Sigma \mathbf{w}$ gives $\mathbf{w} \propto \Sigma^{-1} \boldsymbol{\mu}$

	GMV	Tangency (MV)
Weights	$\mathbf{w}_{\text{GMV}} = \Sigma^{-1} \mathbf{1} / A$	$\mathbf{w}_{\text{MV}} = \Sigma^{-1} \boldsymbol{\mu} / B$, $B = \mathbf{1}^\top \Sigma^{-1} \boldsymbol{\mu}$
Result	variance $\sigma_{\text{GMV}}^2 = 1/A$	squared Sharpe $\theta^2 = \boldsymbol{\mu}^\top \Sigma^{-1} \boldsymbol{\mu}$
Needs	Σ only	$\boldsymbol{\mu}$ and Σ ; Sharpe $\text{sign}(B) \theta$

- The normalised $\Sigma^{-1} \boldsymbol{\mu} / B$ is the maximum only if $B > 0$; $B < 0$ gives the minimum-Sharpe portfolio, $B = 0$ no fully invested solution

Separation and the Frontier in Closed Form

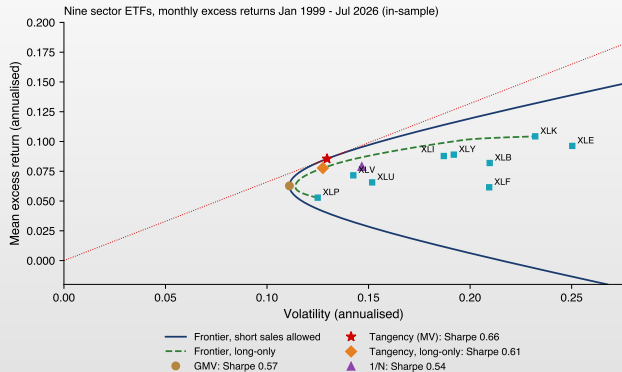


James Tobin, Nobel Prize in
Economics 1981

Photo: unknown author (1962); public
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- Tobin (1958): add a riskless asset paying R_f ; every investor holds the **same** risky portfolio, the tangency portfolio, mixed with cash or borrowing
 - ▶ Risk aversion changes only the mix along the capital allocation line: **two-fund separation**, the bridge to the CAPM (Chapter 3)
- Merton (1972): with short sales, the frontier is a hyperbola; Σ positive definite, μ not proportional to 1
 - ▶ $A = 1^\top \Sigma^{-1} 1$, $B = 1^\top \Sigma^{-1} \mu$, $C = \mu^\top \Sigma^{-1} \mu$, $D = AC - B^2 > 0$
$$\sigma^2(m) = \frac{Am^2 - 2Bm + C}{D}$$
- The GMV sits at $m = B/A$, $\sigma^2 = 1/A$; every frontier portfolio mixes any two distinct frontier portfolios, e.g. GMV and tangency (if $B \neq 0$)

The Frontier of US Sectors



- Monthly excess returns, January 1999 – July 2026 ($T = 331$); all parameters estimated on the full sample: an **in-sample** picture
- Tangency: Sharpe 0.66; GMV: 0.57 (volatility 11.1%); $1/N$: 0.54
- Tangency weights: -0.37 financials, -0.21 materials, $+0.37$ health care: short positions
- Long-only tangency: Sharpe 0.61; materials, financials and industrials get exactly zero, a **corner solution**
- The gain over $1/N$ is small even with perfect hindsight

Constraints

- ▣ **Long-only (LO)**: $w_i \geq 0$; no closed form, solved by quadratic programming
- ▣ Position limits: $w_i \leq 0.2$; sector or asset-class limits: $\sum_{i \in G} w_i \leq c_G$
- ▣ Leverage limit: $\sum_i |w_i| \leq L$
- ▣ **Frost & Savarino (1988)**: constraining weights improved out-of-sample performance; why this works is a shrinkage result (next section)
- ▣ MV-LO and GMV-LO are the long-only versions of MV and GMV

Means Are Hard to Estimate: Merton's Argument

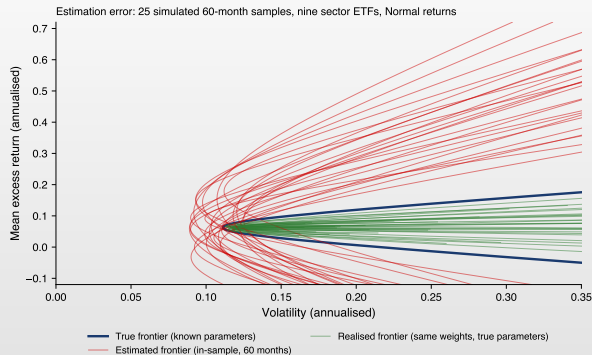


Robert C. Merton, Nobel Prize in
Economics 1997

Photo: Digarnick (2006); public
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- Merton (1980): Y years sampled h times a year, i.i.d. Normal returns (a diffusion) with annual mean μ and variance σ^2
 - ▶ $\text{Var}(\hat{\mu}) = \sigma^2/Y$: independent of h ; only the **span** of the sample helps
 - ▶ $\text{Var}(\hat{\sigma}^2) \approx 2\sigma^4/(hY) \rightarrow 0$ as $h \rightarrow \infty$ (Normal; in general $(\kappa - 1)\sigma^4/(hY)$, κ the kurtosis): variances sharpen with frequency
- SPY, $Y = 24$, $\sigma = 14.7\%$: standard error 3.0%, 95% interval [5.8%, 17.6%] for a mean of 11.7%
- A standard error of 1% needs $(0.147/0.01)^2 \approx 217$ years
- Chopra & Ziemba (1993): errors in means cost about ten times more than errors in variances; rules without means (GMV, risk parity) are more robust

Error Maximisation: Estimated versus Realised Frontiers



- **Michaud (1989):** optimisers are **estimation-error maximisers**; **Best & Grauer (1991):** a tiny change in one mean moves many weights
- True parameters = sector moments 1999–2026; draw 60 Normal months, optimise, evaluate with the true parameters
- Red: estimated frontiers, too optimistic; green: the same portfolios under the true parameters, all below the true frontier
- 2000 simulated tangency portfolios: median estimated maximum Sharpe 1.56, median true Sharpe 0.30, true optimum 0.66
- 96% have a true Sharpe below $1/N$ (0.54), 7.5% below zero; in 11% of samples $1^\top \hat{\Sigma}^{-1} \hat{\mu} < 0$

The Plug-in Sharpe Ratio Is Biased

- i.i.d. $N(\mu, \Sigma)$ returns, maximum-likelihood $\hat{\mu}$, $\hat{\Sigma}$, $\hat{\theta}^2 = \hat{\mu}^\top \hat{\Sigma}^{-1} \hat{\mu}$ (Kan & Zhou (2007); Wishart moments in the [Appendix](#))

$$\mathbb{E}[\hat{\theta}^2] = \frac{T}{T - N - 2} \left(\theta^2 + \frac{N}{T} \right) = \frac{T\theta^2 + N}{T - N - 2}, \quad T > N + 2$$

- Sectors: $N = 9$, $T = 60$, true $\theta = 0.66$ a year: $\mathbb{E}[\hat{\theta}^2]$ corresponds to a Sharpe of 1.65 a year, the analytical counterpart of the simulated median 1.56

- Unbiased estimator of θ^2 : $\hat{\theta}_u^2 = [(T - N - 2)\hat{\theta}^2 - N] / T$

► It can be negative; truncating at zero gives a nonnegative but upward-biased estimator, and $\sqrt{\hat{\theta}_u^2}$ is not unbiased for θ

- Sectors, full sample ($T = 331$): plug-in $\hat{\theta} = 0.66$, bias-corrected $\sqrt{\hat{\theta}_u^2} = 0.31$ a year: more than half of the in-sample Sharpe is noise, even with 27 years

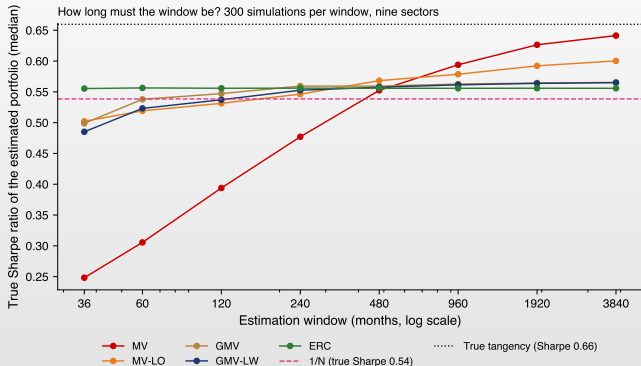
- The tangency weights are OLS coefficients of a regression of 1 on the returns (Britten-Jones (1999); [Appendix](#)), Seminar A2 and B9)

Unstable Weights



- Nine sector ETFs; each month the weights use only the previous 60 months; top panel clipped at ± 6
- MV: median gross exposure $\sum_i |w_i| = 4.9$, above 3 in 90% of the months; largest single weight 26 in absolute value
- Explosions in 2004–2006 and 2009: the denominator $1^\top \hat{\Sigma}^{-1} \hat{\mu}$ is close to zero, so the tangency weights have very heavy tails
 - Negative in 12 of 271 windows: MV then holds the minimum-Sharpe portfolio
- GMV-LW (bottom): gross exposure 1.5, largest weight 0.94
- Unstable weights mean high turnover, hence high trading costs

How Long Must the Window Be to Beat 1/N?



- DeMiguel, Garlappi & Uppal (2009): 14 models on 7 data sets; none beats 1/N consistently in Sharpe ratio, certainty-equivalent return or turnover; their calibration needs about 3000 months for 25 assets
- Here: 300 simulated samples per window, population parameters = sector moments
- MV beats 1/N only with 480 months (40 years): median true Sharpe 0.55 vs 0.54; with 60 months 0.31
- MV-LO, GMV, GMV-LW are close to 1/N from 60 months on; ERC is almost insensitive, 0.56
- In this calibration 1/N is nearly efficient (true tangency 0.66): simulation evidence, not an empirical test

 MFM_ch4_window_length

Why the Sample Covariance Matrix Fails

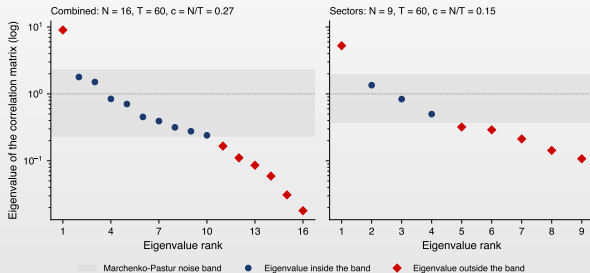
- ▣ Sample eigenvalues are too dispersed: the largest too large, the smallest too small
- ▣ **Condition number**: largest over smallest eigenvalue; a large value means Σ^{-1} amplifies noise
- ▣ Optimisers invert Σ , so they load on the directions with the smallest estimated variance
- ▣ The demeaned sample matrix has rank at most $T - 1$: singular, not invertible, whenever $T \leq N$
- ▣ **Marchenko–Pastur** benchmark ([Marčenko & Pastur \(1967\)](#)): i.i.d. returns with $\Sigma = \sigma^2 I$; as $N, T \rightarrow \infty$ with $N/T \rightarrow c \in (0, 1)$, the sample eigenvalues fill

$$[\lambda_-, \lambda_+] = [\sigma^2(1 - \sqrt{c})^2, \sigma^2(1 + \sqrt{c})^2]$$

- ▶ $c = 16/60 = 0.27$: band $[0.23, 2.30]$ for a correlation matrix ($\sigma^2 = 1$), a ten-fold spread from noise alone
- ▶ [Laloux et al. \(1999\)](#): eigenvalues outside the band are candidates for structure; those inside cannot be told from noise by this comparison alone

Sample Eigenvalues versus the Noise Band

Last 60-month window, Aug 2021 - Jul 2026



- Correlation matrices of monthly excess returns, last 60-month window (August 2021 – July 2026); shaded band: $[\lambda_-, \lambda_+]$ for $\sigma^2 = 1$
- Only the market mode lies above λ_+ : 9.0 (56% of the trace) for 16 ETFs, 5.2 for nine sectors ($\lambda_+ = 1.92$); with the noise variance rescaled to $\sigma^2 = 1 - \lambda_1/N$ (Laloux et al. (1999)), three eigenvalues exceed the band in each universe
- Six of 16 eigenvalues lie **below** the band: near-riskless combinations such as TLT against IEF; GMV loads exactly on them, so if they are underestimated, so is its risk

The Shrinkage Estimator

- Shrink the sample matrix S towards a structured target F

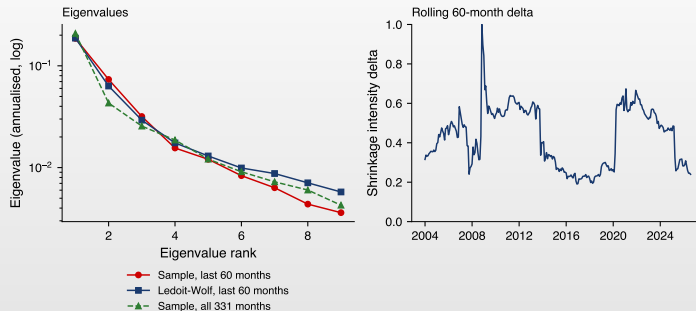
$$\hat{\Sigma}_{\text{LW}} = \delta F + (1 - \delta) S, \quad 0 \leq \delta \leq 1$$

- Targets

- ▶ Single-index (market) model: Ledoit & Wolf (2003)
- ▶ Scaled identity: Ledoit & Wolf (2004a)
- ▶ Constant correlation, $f_{ii} = s_{ii}$ and $f_{ij} = \bar{r} \sqrt{s_{ii} s_{jj}}$ for $i \neq j$, $\bar{r}$ the average sample correlation: Ledoit & Wolf (2004b), used here

- δ trades the variance of S against the bias of F : noisy data or a good target, large δ ; long samples, $\delta \rightarrow 0$ (the optimal δ^* in the [Appendix](#))

Shrinkage on US Sectors



- Nine sector ETFs, monthly excess returns; last window August 2021 – July 2026: $\delta = 0.24$; the smallest eigenvalues are pulled up, the largest almost unchanged; condition number $51.6 \rightarrow 32.5$
- Rolling δ : mean 0.43, range 0.19–1.00; $\delta = 1$ in the early windows, where 60 months are too few to beat the target
- GMV-LW: the GMV portfolio computed with $\hat{\Sigma}_{LW}$

Constraints Are Shrinkage Too

- ▣ Jagannathan & Ma (2003): the long-only GMV equals the unconstrained GMV of a modified covariance matrix

$$\tilde{S} = S - (\lambda \mathbf{1}^\top + \mathbf{1} \lambda^\top), \quad \mathcal{L} = \frac{1}{2} \mathbf{w}^\top S \mathbf{w} - \lambda_0 (\mathbf{w}^\top \mathbf{1} - 1) - \lambda^\top \mathbf{w}$$

- ▶ $\lambda \geq 0$: the Lagrange multipliers of the constraints $w_i \geq 0$ in $\mathcal{L}$; $\lambda_i > 0$ only where the constraint binds
- ▶ The first-order condition $S \mathbf{w} = \lambda_0 \mathbf{1} + \lambda$ and $\lambda^\top \mathbf{w} = 0$ give $\tilde{S} \mathbf{w} = \lambda_0 \mathbf{1}$: the GMV condition for $\tilde{S}$
- ▣ $\tilde{s}_{ij} = s_{ij} - \lambda_i - \lambda_j$: a binding constraint on asset i lowers its covariances, which shrinks the large estimates
- ▣ Imposing the “wrong” constraint (no short sales) can reduce out-of-sample risk; the effect is largest when covariances come from few observations

Nonlinear Shrinkage

- Linear shrinkage mixes two matrices with one δ ; **nonlinear** shrinkage keeps the sample eigenvectors and corrects each eigenvalue separately
 - ▶ Ledoit & Wolf (2017): tuned for portfolio selection; Ledoit & Wolf (2020): an analytical formula, no optimisation
- Out of sample, 60-month window, GMV volatility
 - ▶ Sectors: sample 11.90%, linear (constant correlation) 11.80%, nonlinear 11.62%
 - ▶ 16 ETFs: sample 4.06%, linear 4.38%, nonlinear 4.31%: better than the linear target, not than the raw matrix
- A constant correlation fits neither stock–stock (positive) nor stock–bond (negative) pairs: the target matters (Seminar 4, B4)
- Rule: shrink, constrain, or both, but never invert a raw sample matrix of many assets

From Covariance Shrinkage to Mean Shrinkage



Fischer Black (1938–1995)

Photo: Dalmatine at English

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[Commons](#)

- Shrinkage fixes $\hat{\Sigma}$; Black & Litterman (1992), developed at Goldman Sachs, shrinks the means towards those implied by a **benchmark** portfolio w_b
- Reverse optimisation: a mean–variance investor with risk aversion $\delta_{\mathbf{RA}}$ and access to cash holds $w^* = (\delta_{\mathbf{RA}} \Sigma)^{-1} \mu$ in risky assets; the returns that make w_b optimal are

$$\pi = \delta_{\mathbf{RA}} \Sigma w_b$$

- π : **implied** (equilibrium) excess returns; with market weights, the CAPM returns
- Here: no market capitalisations for the sectors, so $w_b = 1/N$, $\delta_{\mathbf{RA}} = 2.5$
- Then add the investor's **views**, weighted by their confidence: the weights move away from the benchmark only where views exist

Views and the Posterior

▣ K views: $P\mu = q + \varepsilon$, $\varepsilon \sim N(0, \Omega)$; prior $\mu \sim N(\pi, \tau\Sigma)$

▣ Posterior mean: a precision-weighted average; weights $w_{BL} = (\delta_{RA}\Sigma)^{-1}\mu_{BL}$

$$\mu_{BL} = [(\tau\Sigma)^{-1} + P^\top \Omega^{-1}P]^{-1}[(\tau\Sigma)^{-1}\pi + P^\top \Omega^{-1}q]$$

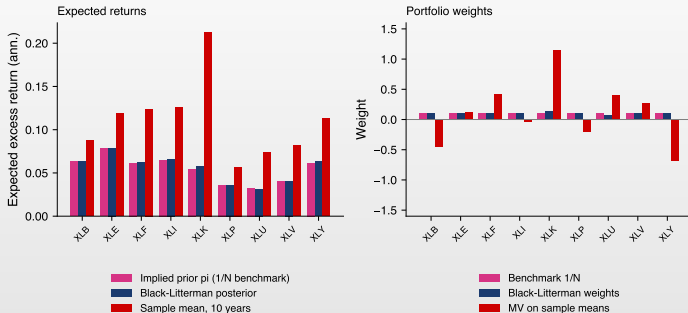
▣ Limits: $\Omega \rightarrow 0$ gives $P\mu_{BL} = q$ exactly; $\Omega \rightarrow \infty$ gives π ; without views, the benchmark itself

▣ He & Litterman (2002): $\tau = 0.05$, $\Omega = \text{diag}(P\tau\Sigma P^\top)$; with Σ in the weights, only the assets in the views move

▣ One asset, $\pi = 5\%$, $\tau\sigma^2 = 0.00128$, view 8% with variance 0.0004: $\mu_{BL} = 7.29\%$, view weight 0.76

▣ The same estimator is a GLS (mixed) regression (Theil & Goldberger (1961); [Appendix](#), Seminar A5)

Black–Litterman on US Sectors



- Σ from monthly excess returns, August 2016 – July 2026; one view: XLK minus XLU = 3% a year; implied 2.2%, posterior 2.6% (the average, since Ω equals the prior variance of the view)
- Weights: XLK from 11.1% to 14.4%, XLU from 11.1% to 7.8%, all others unchanged; the view is relative ($P1 = 0$), so the tilt $\propto P^\top$ sums to zero and the portfolio stays fully invested
- Sample means instead: XLK 21.3% a year; MV weights +115% XLK, -68% XLY, -46% XLB

Risk Contributions

- Volatility is homogeneous of degree one in the weights (Euler)

$$\sigma_p = \sum_{i=1}^N w_i \frac{\partial \sigma_p}{\partial w_i} = \sum_{i=1}^N \frac{w_i (\Sigma w)_i}{\sigma_p}$$

- Risk contribution** of asset i : $RC_i = w_i (\Sigma w)_i / \sigma_p$; percentage share $RC_i / \sigma_p = w_i (\Sigma w)_i / \sigma_p^2$
- Worked example, 50/50 in SPY and IEF (monthly 2002–2026, annualised): $\sigma_1^2 = 0.02166$, $\sigma_2^2 = 0.00441$, $\sigma_{12} = -0.00098$
 - $(\Sigma w)_1 = 0.5(0.02166) + 0.5(-0.00098) = 0.01034$, $(\Sigma w)_2 = 0.5(-0.00098) + 0.5(0.00441) = 0.00172$
 - $\sigma_p^2 = 0.5(0.01034) + 0.5(0.00172) = 0.00603$, $\sigma_p = 7.76\%$; shares: SPY 85.8%, IEF 14.2%
- Equal weights are not equal risks: volatile assets dominate; risk budgeting chooses w so that the shares equal preset budgets b_i

Equal Risk Contribution

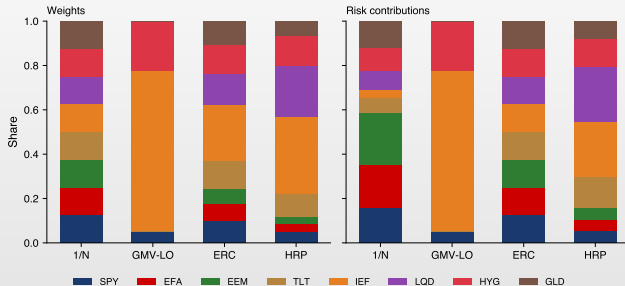
- ERC (Maillard, Roncalli & Teïletche (2010)): $RC_i = RC_j$ for all i, j ; computed from a convex problem, then normalised

$$\min_{y>0} \frac{1}{2} y^\top \Sigma y - \frac{1}{N} \sum_i \ln y_i, \quad w = y / 1^\top y$$

- For Σ positive definite a solution exists and is unique (strictly convex objective, infinite at the boundary)
- Volatility ordering: $\sigma_{\text{GMV}} \leq \sigma_{\text{ERC}} \leq \sigma_{1/N}$
- Two assets: ERC is **inverse volatility**, $w_1 = \sigma_1^{-1} / (\sigma_1^{-1} + \sigma_2^{-1})$, for any $\rho \in (-1, 1)$
 - SPY (14.7%) and IEF (6.6%): ERC holds 31% SPY, volatility 6.1%, each asset 50% of the variance
- Needs no expected returns, only Σ ; ERC is the tangency portfolio if all assets have the same Sharpe ratio and equal correlations

Weights versus Risk Contributions

Eight multi-asset ETFs, monthly May 2007 - Jul 2026

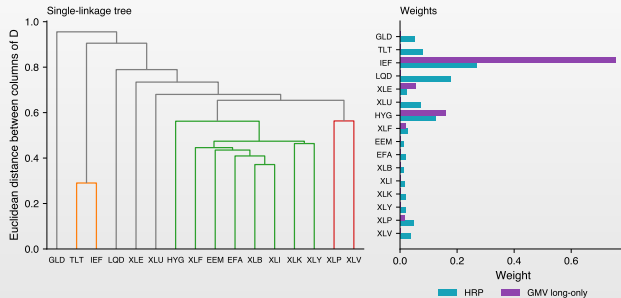


- Eight multi-asset ETFs, full-sample covariance matrix, May 2007 – July 2026 ($T = 231$)
- 1/N: equities and high yield receive half of the weight but 69% of the risk; ERC: 12.5% of the risk from every asset, IEF gets the largest weight (25%); long-only GMV: 73% IEF, 22% HYG, 5% SPY, five assets at zero
- Volatility: 1/N 9.4%, ERC 7.9%, HRP 7.1%, long-only GMV 5.8%: ERC lies between 1/N and GMV, as the theory predicts

Hierarchical Risk Parity: From a Tree to Weights

- HRP (López de Prado (2016)), three steps, no matrix inversion (works even when Σ is singular), all weights positive
 1. **Tree**: correlation distance $d_{ij} = \sqrt{(1 - \rho_{ij})/2} \in [0, 1]$; then the Euclidean distance between the **columns** of $D = (d_{ij})$, $\tilde{d}_{ij} = \sqrt{\sum_k (d_{ki} - d_{kj})^2}$; single linkage on $\tilde{d}_{ij}$ (repeatedly merge the two closest clusters) gives a **dendrogram**
 2. **Quasi-diagonalisation**: order the assets as the leaves of the tree, so that similar ones are adjacent
 3. **Recursive bisection**: split the **ordered list** in two halves; give each half weight in inverse proportion to its variance $\tilde{w}^\top \Sigma \tilde{w}$, with $\tilde{w}$ the inverse-variance weights inside the half; repeat inside each half
- Mantegna (1999): the same distance and the **minimum spanning tree** reveal the hierarchy of a stock market; textbook: Härdle & Simar (2019), cluster analysis
- A worked four-asset example: Seminar 4, B3

HRP on 16 ETFs



- Nine sectors and seven multi-asset ETFs, monthly, May 2007 – July 2026; closest pair TLT and IEF ($\tilde{d} = 0.29$), gold joins last (0.96)
- Equity cluster: cyclical sectors, EFA, EEM and high-yield credit; defensive sectors (utilities, staples, health care) and energy join it late
- HRP: all 16 weights positive, largest IEF 26.7%, LQD 17.6%, HYG 12.6%; long-only GMV: 75.6% in IEF and only five assets above zero

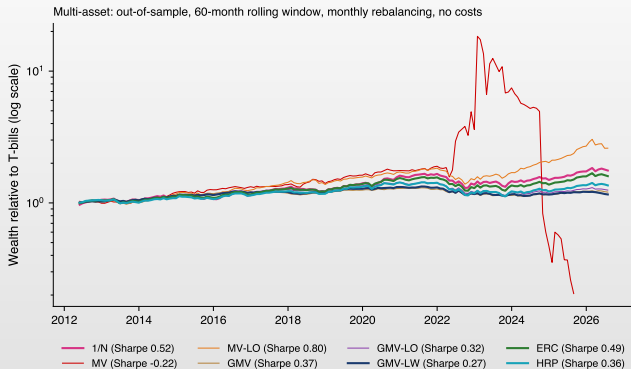
Backtest Design

- ▣ Rolling window of 60 months, monthly rebalancing; the weights of month t use months $t - 60, \dots, t - 1$ only
- ▣ Eight rules: 1/N, MV, MV-LO, GMV, GMV-LO, GMV-LW, ERC, HRP
- ▣ Three universes
 - ▶ Sectors: OOS January 2004 – July 2026, 271 months (first window January 1999 – December 2003)
 - ▶ Multi-asset and combined: OOS May 2012 – July 2026, 171 months
- ▣ Annualised mean, volatility and Sharpe ratio of excess returns; maximum drawdown of total-return wealth
- ▣ Every Sharpe ratio difference is reported with a test (next slides), and every test run enters the multiple-testing correction

Turnover and Costs: One Worked Month

- Between rebalancings weights drift with returns: $w_{i,t}^+ = w_{i,t}(1 + R_{i,t}) / \sum_j w_{j,t}(1 + R_{j,t})$
- **Turnover**: traded fraction of wealth, $TO_{t+1} = \sum_i |w_{i,t+1} - w_{i,t}^+|$
- Worked month, sectors, 1/N: in January 2004 XLB returned -4.4% and XLK $+4.3\%$, so the weights of 11.11% drifted to 10.50% and 11.45%
 - ▶ Rebalancing back to 1/N in February 2004 trades $TO = 0.019$: 1/N also trades, selling winners and buying losers
 - ▶ MV-LO, re-estimated on the new window, trades 0.748 in the same month
- Net return with cost c per unit traded: $R_{p,t}^{\text{net}} = R_{p,t} - c TO_t$; 1 basis point (bp) = 0.01%, so 20 bp means $c = 0.002$
 - ▶ At 20 bp the February trade costs MV-LO $0.002 \times 0.748 = 0.15\%$ of wealth, 1/N 0.004%

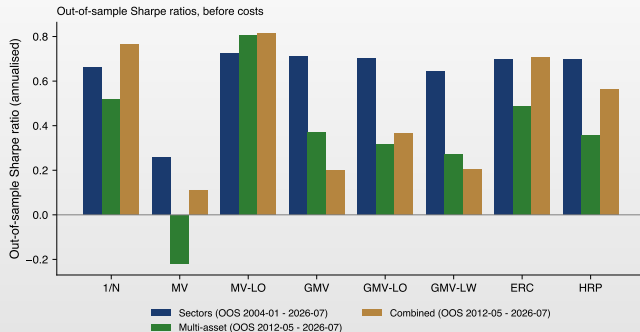
Out-of-Sample Growth: Multi-Asset



- Eight multi-asset ETFs; wealth relative to a Treasury-bill account, before costs
- MV collapses: volatility 518% a year, turnover 44 times wealth per month; a monthly loss above 100% ends its wealth path
- MV-LO has the highest Sharpe (0.80); 1/N 0.52; ERC 0.49
- GMV rules: lowest volatility (3.8%–5.4%), low Sharpe (0.27–0.37); they load on Treasuries, whose excess returns were slightly negative after May 2012 (TLT -0.5%, IEF -0.2% a year)
- Is MV-LO really better, or lucky? A test is needed

 MFM_ch4_oos_backtest

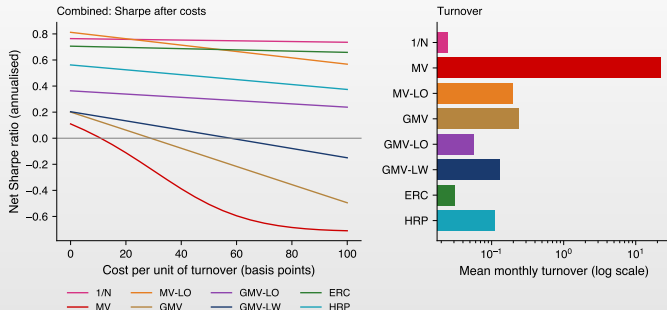
Sharpe Ratios Across Universes



- 60-month rolling window, monthly rebalancing, annualised, before costs; complete tables in the [Appendix](#)
- Sectors: every rule except MV lands within 0.07 of 1/N (0.66); MV 0.26 with drawdown -100%
- Combined: 1/N 0.76, MV-LO 0.81, ERC 0.71; the GMV rules 0.20–0.36
- MV: a monthly loss above 100% wipes out the capital; its mean and Sharpe describe a hypothetical return series, not investable wealth
- Turnover per month, sectors: 1/N 0.02, ERC 0.03, GMV-LW 0.14, MV-LO 0.23, MV 3.86

 [MFM_ch4_oos_backtest](#)

Sharpe Ratios after Costs



- Combined universe (16 ETFs); cost c from 0 to 100 bp per unit of turnover; turnover from 0.02 (1/N) to 22 (MV) per month
- Net Sharpe turns negative at about 11 bp for MV, 29 bp for GMV, 58 bp for GMV-LW; MV-LO overtakes 1/N only for costs below about 22 bp (Seminar 4, B8)
- 1/N and ERC barely move: low turnover is a hidden advantage; report net performance for realistic costs, not only gross

Inference on One Sharpe Ratio: Lo (2002)

- Lo (2002), i.i.d. Normal returns: $\sqrt{T}(\widehat{SR} - SR) \rightarrow N(0, 1 + SR^2/2)$, with SR monthly
- 1/N on sectors, $T = 271$: monthly $\widehat{SR} = 0.191$, standard error 0.061
 - ▶ Annualised $0.66 \pm 1.96 \times 0.21$: 95% interval $[0.25, 1.08]$
- Annualising by $\sqrt{q}$ ($q = 12$) is exact only without autocorrelation; in general $SR(q) = \eta(q) SR$

$$\eta(q) = \frac{q}{\sqrt{q + 2 \sum_{k=1}^{q-1} (q-k) \rho_k}}$$

- Here $\rho_1 = -0.03$, $\sum_{k=1}^{11} \rho_k = -0.11$: $\eta = 3.62$ instead of $\sqrt{12} = 3.46$, annual Sharpe 0.69

Testing a Difference in Sharpe Ratios

- H_0 : $SR_1 = SR_2$ for two strategies evaluated on the same months; [Jobson & Korkie \(1981\)](#): a test under i.i.d. Normal returns
- [Ledoit & Wolf \(2008\)](#): a robust version
 - ▶ $\hat{\Delta} = f(\hat{v})$, $v = (\mu_1, \mu_2, \mathbb{E}[r_1^2], \mathbb{E}[r_2^2])$, $f = \frac{\mu_1}{\sqrt{\mathbb{E}r_1^2 - \mu_1^2}} - \frac{\mu_2}{\sqrt{\mathbb{E}r_2^2 - \mu_2^2}}$
 - ▶ Standard error by the delta method: $\sqrt{\nabla f^\top \hat{\Psi} \nabla f / T}$, $\hat{\Psi}$ a HAC covariance
- Strategies on the same assets are highly correlated: small differences can be significant

The Block Bootstrap

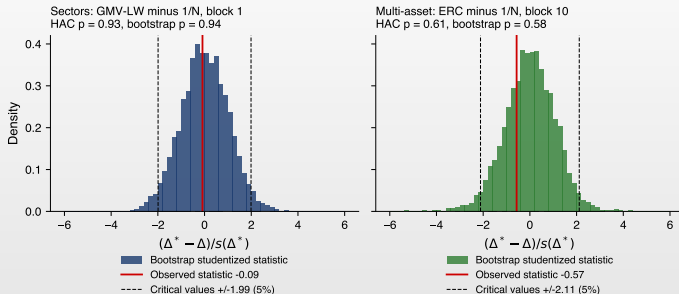


Bradley Efron, inventor of the bootstrap (1979)

Photo: Ryan K. Morris (2007); public domain; Wikimedia Commons

- Bootstrap: resample the data to approximate the sampling distribution of $\hat{\Delta}$
- Returns are dependent (volatility clustering): resample **blocks** of consecutive months (Künsch (1989))
- Ledoit & Wolf (2008), Section 3.2.2: circular blocks of **pairs** (r_{1t}, r_{2t}) , wrapping around the end of the sample, so the correlation of the two strategies is kept
- Centre and studentize: bootstrap $|\hat{\Delta}^* - \hat{\Delta}|/s(\hat{\Delta}^*)$; asymptotically pivotal, more accurate than the Normal approximation of the HAC test
- $p = [\#\{\tilde{d}^{*m} \geq |\hat{\Delta}|/s(\hat{\Delta})\} + 1]/(M + 1)$, $M = 4999$ draws
- Block length chosen by the calibration of Ledoit & Wolf (2008), Algorithm 3.1 (Appendix): mostly 1 for sectors, 8–10 for multi-asset and combined rules

Is Anything Better than $1/N$?



- Bootstrap distributions of the studentized difference against $1/N$; red: observed statistic; dashed: bootstrap critical values
- Sectors, GMV-LW minus $1/N$: -0.02 , 95% interval $[-0.36, 0.33]$, HAC $p = 0.93$, bootstrap $p = 0.94$ (block 1); multi-asset, ERC minus $1/N$: -0.03 , $[-0.15, 0.08]$, HAC $p = 0.61$, bootstrap $p = 0.58$ (block 10): so correlated with $1/N$ that the interval is narrow, yet it covers zero
- Sectors: no rule differs from $1/N$ at 5% (closest ERC, $+0.04$, HAC $p = 0.09$); multi-asset: MV-LO $+0.29$ is **not** significant (HAC $p = 0.12$, bootstrap 0.16)

Many Rules, Many Tests

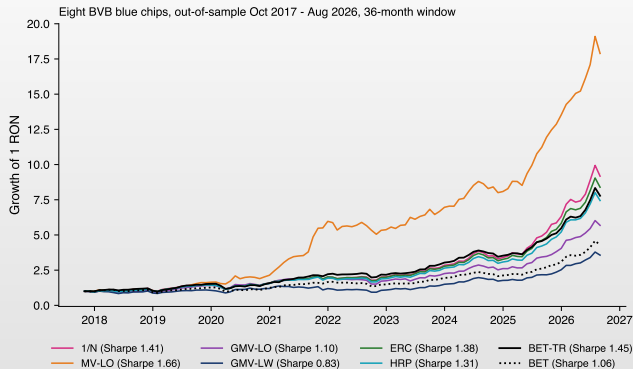
Universe	Rule minus 1/N	$\hat{\Delta}$	HAC p	Bootstrap p
Multi-asset	MV	-0.74	0.01	0.06
Multi-asset	HRP	-0.16	0.07	0.05
Combined	GMV	-0.57	0.03	0.05
Combined	GMV-LO	-0.40	0.05	0.08
Combined	GMV-LW	-0.56	0.02	0.03
Combined	MV-LO	+0.05	0.82	0.82

- No rule beats 1/N significantly in any universe; unadjusted, GMV rules lose to 1/N on the combined universe: low risk, too little return
- The family: $H_0: \Delta_k = SR_k - SR_{1/N} \leq 0$ for all k ; 21 tests (7 rules $\times$ 3 universes); Holm-adjusted p -values are at least 0.29 (HAC) and 0.68 (bootstrap): nothing survives ([Chapter 3](#), multiple testing)
- A different question, the best of 24 configurations: the deflated Sharpe ratio of [Bailey & López de Prado \(2014\)](#) gives $0.84 < 0.95$ for MV-LO on 16 ETFs, but the verdict hinges on the effective number of trials ([Appendix](#))

Setup: a Romanian Investor

- ▣ Eight blue chips listed since 2014: TLV, SNP, BRD, TGN, SNG, SNN, EL, TEL
- ▣ Monthly returns in RON, October 2014 – August 2026; 36-month window, first portfolio October 2017
- ▣ Rules: 1/N, MV-LO, GMV-LO, GMV-LW, ERC, HRP; benchmarks BET-TR and BET
- ▣ Sharpe ratios with $R_f = 0$ (no RON money-market series in the data): return per unit of risk, higher than excess-return Sharpe ratios
- ▣ Earlier Romanian evidence on diversification: [Dragotă, Șerbănescu & Pele \(2008\)](#)

BVB Portfolios versus BET-TR



- Out of sample, October 2017 – August 2026 (107 months), before costs, $R_f = 0$
- BET-TR: mean 24.7% a year, volatility 17.0%, Sharpe 1.45; BET: mean 17.8%, Sharpe 1.07; the gap of about 7% a year is the dividends: compare with BET-TR, never with BET
- 1/N 1.41, ERC 1.38, HRP 1.31: close to BET-TR
- MV-LO 1.66 (difference $+0.21$, $p = 0.44$); GMV-LW 0.83 (-0.63 , $p = 0.05$)
- Average correlation of the eight stocks 0.44: little room for diversification inside the BVB; the latest GMV-LO puts 70% in TLV
- Open question: does diversification **beyond** BET pay? (Seminar 4, C1)

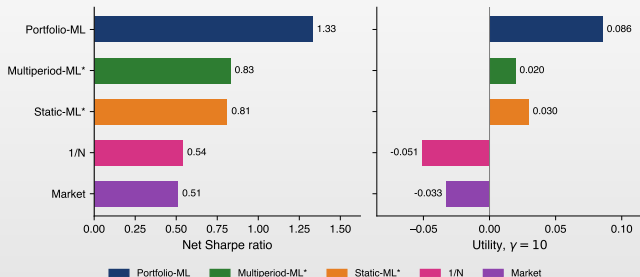
Case Study: Which Frontier Can a Large Investor Reach?

- Jensen, Kelly, Malamud & Pedersen (2026), *Review of Financial Studies*: how should portfolios be built from machine-learning return forecasts when every trade moves prices?
- **Implementable efficient frontier** (eq. 12): for each volatility σ , the highest expected return **net of trading costs**, $k(\sigma) = \max_{\pi} E[r_{t+1}^{\text{net}}]$ s.t. $E[\pi_t^\top \Sigma \pi_t] = \sigma^2$
 - Proposition 1: the net Sharpe ratio $k(\sigma)/\sigma$ falls along the frontier, and a larger investor faces a lower frontier
- Cost as a fraction of wealth W_t (eqs. 4, 7): $c_t = \frac{W_t}{2}(\pi_t - g_t \pi_{t-1})^\top \Lambda (\pi_t - g_t \pi_{t-1})$, quadratic in the trade; g_t drifts last month's weights, $\Lambda = \text{diag}(0.2/V_i)$, V_i the dollar volume
- Optimal rule: trade only part of the way towards an **aim portfolio**, $\pi_t = m g_t \pi_{t-1} + (1 - m)A_t$ (Gârleanu & Pedersen (2013)); Portfolio-ML learns A_t from the signals by a ridge regression that maximises utility net of costs

	Paper	Course replication
Assets	NYSE stocks above median size (184–805)	14 US stocks + 9 sector ETFs
Signals	115 characteristics (Jensen, Kelly & Pedersen (2023))	5 price and volume signals
Out of sample	1981–2020, monthly	February 2014 – August 2026 (151 months)
Investor	$\gamma = 10$, wealth \$10 billion at the end of 2020	same

Case Study: Table 2 at a Glance

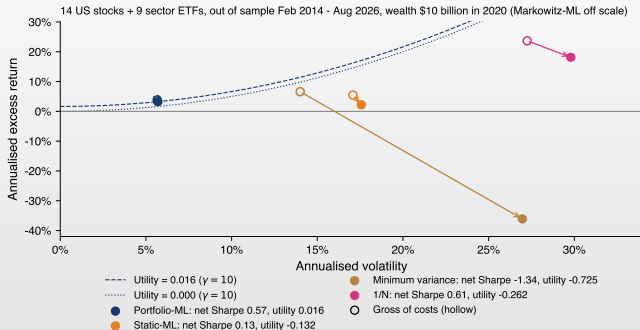
Out of sample 1981–2020, US stocks, wealth \$10 billion (Markowitz-ML off scale)



- Jensen, Kelly, Malamud & Pedersen (2026), Table 2: net of costs, 1981–2020, $\gamma = 10$; utility (eq. 11) = annual net return $-\frac{\gamma}{2} \times$ annual variance
- Markowitz-ML (mean–variance on ML return forecasts, no costs): gross Sharpe 2.06, turnover 79.36, leverage 74.61: net return extremely negative
- Portfolio-ML: net Sharpe 1.33, utility 0.086 (net return 15%, volatility 11%, cost 0.6% a year)
- Static-ML* (one-period costs; * = extra tuning layer) 0.81, 0.030; 1/N 0.54, -0.051 ; market 0.51, -0.033 : only cost-aware rules have positive utility

 MFM_ch4_implementable_frontier

Case Study: Gross and Net of Costs on the Course Data



- Hollow marker: before costs; filled: after the costs of eq. (35); dashed curve: pairs with the utility of Portfolio-ML (0.016), dotted: utility 0
- Portfolio-ML barely moves: 3.8% gross, 3.3% net, 5.7% volatility; net Sharpe 0.57
- 1/N keeps 18.1% net (Sharpe 0.61), but at 29.8% volatility its utility is -0.26
- Minimum variance: +6.6% gross, -36.1% net, rebalancing a leverage of 2.6 costs 43% a year
- Limits: 23 assets, 5 signals, no factor model for Σ ; the frontier that matters is the one after costs (Seminar 4, C3)

 MFM_ch4_implementable_frontier

Open Questions

- Questions
 - ▶ does any **estimated** allocation rule beat $1/N$ out of sample, net of costs?
 - ▶ does it still beat $1/N$ once every rule, window and universe tried is counted?
- Live evidence points both ways
 - ▶ Nonlinear shrinkage of Σ improves GMV out of sample in large cross-sections: [Ledoit & Wolf \(2017\)](#)
 - ▶ Combining tangency, GMV and $1/N$ under estimation risk: [Kan, Wang & Zhou \(2022\)](#)
 - ▶ Backtests need haircuts for the number of specifications tried: [Harvey & Liu \(2015\)](#)
- Formal statement for K rules on one universe: $H_0: \Delta_k = SR_k - SR_{1/N} \leq 0$ for all k ; $H_1: \Delta_k > 0$ for some k ; error rate fixed for the whole family
- One pass of the loop with an AI assistant: literature with DOIs, competing hypotheses, code on the course's monthly data, a pre-registered family of tests, a referee-style critique (prompts in the [Appendix](#))
- Falsified by: a positive $\hat{\Delta}_k$ that survives a family-wise correction, costs and a second sample

What the Researcher Must Check

- **Look-ahead:** weights at t built with month- t returns; assert that the last estimation date is before t
- **Survivorship and selection:** an ETF list chosen today; use the universe available at each date, including later delisted ETFs, or disclose the limitation
- **Multiple testing:** 7 rules $\times$ 3 universes already give 21 tests; Holm (FWER) or Benjamini–Hochberg (FDR, [Benjamini & Hochberg \(1995\)](#)) adjusted p -values
- **Power:** $\hat{\Delta}$ has a standard error near 0.18 on 171 months; compute the detectable effect before running
- **Return type:** portfolios aggregate **simple** returns, Sharpe ratios use **excess** returns; recompute one month step by step
- **Invented references:** an unresolvable DOI or a title mismatch drops the claim and goes into `AI_ERRORS.md`

Mini-Case: Data Snooping in This Chapter's Backtest

- ▣ A typical assistant proposal: “MV-LO beats $1/N$ on the multi-asset universe”; superiority is one-sided: $H_0: \Delta \leq 0$, $p = 1 - \Phi(\hat{\Delta}/\text{se})$
- ▣ Replicate: $\hat{\Delta} = +0.29$, HAC standard error 0.18, one-sided $p = 0.06$ (two-sided HAC 0.12, bootstrap 0.16)
- ▣ Test the family, not the winner: 21 tests (7 rules $\times$ 3 universes)
 - ▶ Only 7 of 21 differences are positive; the 4 two-sided HAC p -values below 0.05 are all **losses**, which never reject a superiority null
 - ▶ One-sided: smallest $p = 0.046$ (sectors, ERC); adjusted Holm 0.96, Benjamini–Hochberg 0.62; 0 discoveries at an FDR of 10%
- ▣ Power 0.8 for $\Delta = 0.29$, one-sided: about 435 months (36 years) at 5%, 944 months (79 years) at the Holm level 5%/21
- ▣ Interpretation: the data cannot rank these rules; the bootstrap interval for MV-LO, $[-0.13, 0.70]$, admits losses and large gains; failure to reject is not proof of equality

Your Turn: A Research Project Idea

▣ Questions

- ▶ does nonlinear shrinkage (Ledoit & Wolf (2017)) beat $1/N$ on the 16 ETFs of this chapter, after 10–50 bp costs?
- ▶ does a three-fund combination (Kan, Wang & Zhou (2022)) beat $1/N$ on the same ETFs, after the same costs?

- ▣ **Replicate first:** the combined-universe results ($1/N$ Sharpe 0.76, MV-LO 0.81) and the adjusted p -values of the mini-case

- ▣ **Pre-register:** hypotheses, family of tests, error rate, power and a placebo (random long-only weights with the same turnover), fixed before any result

- ▣ **Report:** code on the course's monthly data, a table of all specifications, a referee-style self-critique

- ▣ **Log:** prompts in AI_USE.md; every error caught and the test that caught it in AI_ERRORS.md

Conclusions: Which Rule, When?

Rule	Objective	Inputs	Main failure mode	Evidence in this chapter
MV	maximum Sharpe	$\hat{\mu}, \hat{\Sigma}^{-1}$	error maximisation, explosive weights	worst in every universe; insolvent path
MV-LO	maximum Sharpe, $w_i \geq 0$	$\hat{\mu}, \hat{\Sigma}$	corner solutions, turnover	best point estimates, never significant
GMV, GMV-LW	minimum variance	$\hat{\Sigma}$ (shrunk)	loads on low-return assets	lowest risk, loses to 1/N on 16 ETFs (unadjusted)
ERC, HRP	balanced risk	$\hat{\Sigma}$	a tilt of 1/N; HRP trades more	never significantly better than 1/N
Black–Litterman	benchmark plus views	w_b , views, Ω	confidence set too high	moves only the assets in the views
1/N	none	none	ignores risk differences	hard to beat after costs

- Evaluate every rule out of sample, net of costs, with a test of the Sharpe difference and a count of all tests; these verdicts are specific to these samples

Check Yourself, and Next

Retrieval questions

- Why does the in-sample tangency Sharpe overstate the achievable one?
- By how much does it overstate it for $N = 9$, $T = 60$?
- Why does a long-only constraint act like shrinkage?
- Why can a small Sharpe difference between ERC and $1/N$ be nearly significant while a larger one between GMV and $1/N$ is not?

Links to other chapters

- **Chapter 3**: the tangency portfolio and the CAPM; factor covariance models
- **Chapters 5–6**: time-varying volatility and correlation for dynamic weights
- **Chapter 7**: portfolio Value-at-Risk and Expected Shortfall
- **Chapter 8**: backtesting without overfitting
- **Chapter 13**: machine learning for returns and portfolios

Appendix: Wishart Moments and the Distribution of Weights

- i.i.d. $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ returns, maximum-likelihood estimators: $\hat{\boldsymbol{\mu}} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}/T)$, $T\hat{\boldsymbol{\Sigma}} \sim W_N(T-1, \boldsymbol{\Sigma})$ (Wishart), independent
- Inverse Wishart moment, finite for $T > N + 2$: $\mathbb{E}[\hat{\boldsymbol{\Sigma}}^{-1}] = \frac{T}{T-N-2}\boldsymbol{\Sigma}^{-1}$, an **upward** bias that grows with N/T
- Conditioning on $\hat{\boldsymbol{\mu}}$ and using $\mathbb{E}[\mathbf{x}^\top \mathbf{A} \mathbf{x}] = \text{tr}(\mathbf{A} \text{Var}(\mathbf{x})) + \mathbb{E}[\mathbf{x}]^\top \mathbf{A} \mathbb{E}[\mathbf{x}]$ gives $\mathbb{E}[\hat{\theta}^2]$ of the lecture (Seminar A9)
- GMV weights with Wishart $\hat{\boldsymbol{\Sigma}}$ and T large enough relative to N : $\hat{\mathbf{w}}_{\text{GMV}}$ is unbiased and its exact distribution and covariance are known (Okhrin & Schmid (2006))
- Tangency weights divide by $\mathbf{1}^\top \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\mu}}$, which can be near zero: very heavy tails
- Estimation-risk-optimal mixtures of tangency, GMV and $1/N$: Kan & Zhou (2007), Kan, Wang & Zhou (2022)

Appendix: Tangency Weights by Regression, Britten-Jones (1999)

- Britten-Jones (1999): regress the constant 1 on the excess returns, **without** a constant term

$$1 = \mathbf{b}^\top \mathbf{r}_t + u_t, \quad \hat{\mathbf{b}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{1}_T \propto \hat{\Sigma}^{-1} \hat{\boldsymbol{\mu}}$$

- Sherman–Morrison on $\mathbf{X}^\top \mathbf{X} / T = \hat{\Sigma} + \hat{\boldsymbol{\mu}} \hat{\boldsymbol{\mu}}^\top$ gives the proportionality (Seminar A2); $\hat{\mathbf{b}} / \mathbf{1}^\top \hat{\mathbf{b}} =$ sample tangency weights if $\mathbf{1}^\top \hat{\mathbf{b}} > 0$
- OLS t -test of $b_i = 0$: asset i has zero weight in the tangency portfolio; F -test of $\mathbf{R} \mathbf{b} = 0$ with $\mathbf{R} = [\mathbf{I}_{N-1}, -\mathbf{1}_{N-1}]$: “the tangency portfolio is $1/N$ ”
- Exact finite-sample t and F distributions under i.i.d. Normal returns; HAC standard errors otherwise
- Nine sectors, January 1999 – July 2026 ($T = 331$): normalised $\hat{\mathbf{b}}$ reproduces the tangency weights; largest $|t| = 1.12$ (financials), HAC 1.19
- H_0 : tangency = $1/N$, eight restrictions: $F(8, 322) = 0.48$, $p = 0.87$; HAC Wald $\chi_8^2 = 4.50$, $p = 0.81$; not rejecting is not proof of equal weights

Appendix: The Optimal Shrinkage Intensity

- Ledoit–Wolf choose δ to minimise the expected squared (Frobenius) distance to the true Σ

$$\delta^* = \max \left\{ 0, \min \left\{ 1, \frac{\hat{\kappa}}{T} \right\} \right\}, \quad \hat{\kappa} = \frac{\hat{\pi} - \hat{\rho}}{\hat{\gamma}}$$

- $\hat{\pi}$: sum of the asymptotic variances of the entries of S (noise)
- $\hat{\rho}$: their asymptotic covariances with the target entries
- $\hat{\gamma} = \|F - S\|^2$: misspecification of the target
- Complete formulas for the constant-correlation target: [Ledoit & Wolf \(2004b\)](#), Appendix; code: the [shrinkage Quantlet](#)

Appendix: Shrinking the Means and Other Covariance Estimators

- Bayes–Stein ([Jorion \(1986\)](#)): $\hat{\mu}_{BS} = (1 - \hat{\phi}) \hat{\mu} + \hat{\phi} \hat{\mu}_{GMV}$
 - ▶ Empirical-Bayes shrinkage of every mean towards the mean of the GMV portfolio; $\hat{\phi}$ from a Stein argument, larger when N/T is large
- [Tu & Zhou \(2011\)](#): the convex combination of a sophisticated rule and $1/N$ that minimises the expected out-of-sample loss
- [DeMiguel, Garlappi, Nogales & Uppal \(2009\)](#): norm constraints $\|w\|_1 \leq \delta$ or $\|w\|_2 \leq \delta$ nest the long-only GMV, shrinkage GMV and $1/N$; the bridge to the lasso portfolios of [Chapter 13](#)
- Factor structure plus a sparse residual covariance: POET ([Fan, Liao & Mincheva \(2013\)](#)), the link to [Chapter 3](#)
- High-dimensional GMV with shrinkage towards a target portfolio: [Bodnar, Parolya & Schmid \(2018\)](#)

Appendix: Black–Litterman as Mixed (GLS) Estimation

- ▣ Theil & Goldberger (1961): stack the prior and the views as one regression on μ

$$\begin{pmatrix} \pi \\ q \end{pmatrix} = \begin{pmatrix} I \\ P \end{pmatrix} \mu + \begin{pmatrix} e_1 \\ e_2 \end{pmatrix}, \quad \text{Var} \begin{pmatrix} e_1 \\ e_2 \end{pmatrix} = \begin{pmatrix} \tau \Sigma & 0 \\ 0 & \Omega \end{pmatrix}$$

- ▣ GLS gives exactly μ_{BL} , with covariance $M = [(\tau \Sigma)^{-1} + P^\top \Omega^{-1} P]^{-1}$
- ▣ Equivalent form: $\mu_{BL} = \pi + \tau \Sigma P^\top (P \tau \Sigma P^\top + \Omega)^{-1} (q - P \pi)$
- ▣ $\Omega \rightarrow 0$: restricted least squares, $P \mu_{BL} = q$
- ▣ Predictive covariance $\Sigma + M$ in the weights would change all weights; without views it scales them by $1/(1 + \tau)$

Appendix: Other Risk-Based Rules

- ▣ **Inverse volatility:** $w_i \propto 1/\sigma_i$; equals ERC when all correlations are equal
- ▣ **Maximum diversification** (Choueifaty & Coignard (2008)): maximise $w^\top \sigma / \sqrt{w^\top \Sigma w}$, the ratio of the average asset volatility to the portfolio volatility
- ▣ **Minimum variance:** GMV, the extreme risk-based rule
- ▣ All ignore expected returns: robust to estimation error, but optimal only under special assumptions

Appendix: Results, Sector Universe

Rule	Mean	Volatility	Sharpe	Turnover	Max. drawdown
1/N	9.4%	14.3%	0.66	0.02	−49%
MV	17.8%	69.0%	0.26	3.86	−100%
MV-LO	10.7%	14.7%	0.72	0.23	−42%
GMV	8.5%	11.9%	0.71	0.23	−33%
GMV-LO	8.1%	11.5%	0.70	0.09	−34%
GMV-LW	7.6%	11.8%	0.65	0.14	−31%
ERC	9.3%	13.2%	0.70	0.03	−47%
HRP	8.9%	12.8%	0.70	0.08	−45%

- ▣ January 2004 – July 2026; turnover: mean monthly traded fraction of wealth; drawdown of total-return wealth
- ▣ MV: a monthly loss above 100% wipes out the capital (drawdown −100%); its mean and Sharpe describe a hypothetical return series

Appendix: Results, Combined Universe (16 ETFs)

Rule	Mean	Volatility	Sharpe	Turnover	Sharpe, 50 bp
1/N	7.9%	10.3%	0.76	0.02	0.75
MV	28.7%	261.8%	0.11	22.10	-0.51
MV-LO	8.0%	9.8%	0.81	0.20	0.69
GMV	0.8%	4.1%	0.20	0.24	-0.15
GMV-LO	2.0%	5.4%	0.36	0.06	0.30
GMV-LW	0.9%	4.4%	0.20	0.13	0.03
ERC	5.5%	7.8%	0.71	0.03	0.68
HRP	4.0%	7.0%	0.56	0.11	0.47

▣ May 2012 – July 2026; last column: net of 50 basis points per unit of turnover

Appendix: Bootstrap Details and the Block Length

- Studentization: $s(\hat{\Delta}^*)$ from the block sums of each resample; $s(\hat{\Delta})$ on the data from a prewhitened quadratic-spectral kernel ([Andrews & Monahan \(1992\)](#)) with automatic bandwidth ([Andrews \(1991\)](#))
- [Ledoit & Wolf \(2008\)](#), Algorithm 3.1: choose the block b whose bootstrap interval has actual coverage closest to 95%
 - ▶ Fit a first-order vector autoregression to the pair of returns; resample its residuals by a stationary bootstrap ([Politis & Romano \(1994\)](#)), mean block 5
 - ▶ Generate $K = 5000$ pseudo-samples of length T ; in each, build the 95% interval for $b \in \{1, 2, 4, 6, 8, 10\}$
 - ▶ $\hat{g}(b)$: share of intervals that contain the pseudo-true $\hat{\Delta}$; pick $\tilde{b} = \arg \min_b |\hat{g}(b) - 0.95|$
- The grid and K are those of the paper (Sections 4–5); applied to each of the 21 pairs of this chapter
- A non-studentized bootstrap with a fixed 6-month block is not this test: its block is not calibrated for coverage

Appendix: The Deflated Sharpe Ratio

- ▣ Bailey & López de Prado (2014): the best of N Sharpe ratios is biased upwards even if every true Sharpe ratio is zero
- ▣ Expected maximum under H_0 : all true Sharpe ratios zero, N independent trials, monthly units as T ; with $\gamma_E = 0.5772$ (Euler's constant) and $\hat{V}$ the variance of the N estimates

$$SR_0 = \sqrt{\hat{V}} \left[(1 - \gamma_E) \Phi^{-1} \left(1 - \frac{1}{N} \right) + \gamma_E \Phi^{-1} \left(1 - \frac{1}{Ne} \right) \right]$$

- ▣ Deflated Sharpe ratio, with skewness $\hat{\gamma}_3$ and kurtosis $\hat{\gamma}_4$ of the winner

$$DSR = \Phi \left(\frac{(\widehat{SR} - SR_0) \sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2}} \right)$$

- ▣ This chapter: 24 configurations (8 rules $\times$ 3 universes); winner MV-LO on 16 ETFs, 0.81 a year
 - ▶ As 24 independent trials: $SR_0 = 0.52$ a year, $DSR = 0.84 < 0.95$
 - ▶ The configurations are strongly correlated (up to 0.99), so fewer are independent: $N = 8$ gives DSR 0.93, $N = 3$ gives 0.98
- ▣ Holm adjusts the whole family of pairwise tests against $1/N$; the DSR asks whether the single best configuration is better than luck

Appendix: Beyond Equities, Digital Assets

- ▣ [Petukhina, Trimborn, Härdle & Elendner \(2021\)](#): cryptocurrencies in portfolio allocation rules; gains depend on the rule and on liquidity
- ▣ [Pele et al. \(2023\)](#): are cryptos becoming alternative assets?
- ▣ Bitcoin: annualised volatility 172% (monthly, 2010–2026, one surviving coin chosen ex post), IEF 6.6%
- ▣ Risk contributions grow with volatility: risk-based rules give Bitcoin a small weight automatically (Seminar 4, B5)
- ▣ Robust mean–variance for crypto-assets: [Pele \(2023\)](#)

Appendix: Case Study, Method in Detail (1/2)

- Cost (eqs. 4, 7): π_t risky-asset weights, $1 - \mathbf{1}^\top \pi_t$ in the risk-free asset; $g_t = \text{diag}((1 + r_t^f + r_t)/(1 + g_t^w))$ (eq. 6), g_t^w the growth of wealth; $\Lambda = \text{diag}(0.2/V_i)$, V_i the average daily dollar volume over 6 months: trading 1% of V_i moves the price by 0.1% (eq. 35, [Frazzini, Israel & Moskowitz \(2018\)](#))
- Optimal rule (Proposition 3): $\pi_t = m g_t \pi_{t-1} + (1 - m) A_t$, A_t a weighted sum of today's and expected future Markowitz portfolios; m solves eq. (14); [Gârleanu & Pedersen \(2013\)](#) generalised
- Portfolio-ML (Proposition 4, eq. 40): $A_t = \text{diag}(1/\sigma_{i,t}) f(s_t) \beta$, $f = p$ random Fourier features of the signals s_t ; $\hat{\beta} = (E_T[\tilde{\Sigma}_t] + \lambda I)^{-1} E_T[\tilde{r}_{t+1}]$, a ridge regression that maximises utility net of costs in closed form (eqs. 24–26), learning the weights directly as [Brandt, Santa-Clara & Valkanov \(2009\)](#) do without costs

Appendix: Case Study, Method in Detail (2/2)

- Data (Section 5.1): NYSE common stocks above the median market capitalisation (184–805 stocks), 115 characteristics of Jensen, Kelly & Pedersen (2023) ranked into $[0, 1]$; December 1952 – December 2020, out of sample 1981–2020, monthly
- Σ : 13 characteristic themes and 12 industries as factors, exponential weights with half-lives of 378 days (correlations) and 126 days (variances); investor $\gamma = 10$, wealth \$10 billion at the end of 2020, growing with the market (Section 5.1.3)
- Tuning (Table 1): $\lambda \in \{0, e^{-10}, e^{-9.8}, \dots, e^{10}\}$, $p \in \{2^6, \dots, 2^9\}$, $\eta \in \{e^{-3}, e^{-2}\}$ (standard deviation of the random weights), chosen each year by the utility of validation backtests since 1971
- Rivals: Markowitz-ML (eq. 32), Static-ML (one-period cost, eq. 34), Multiperiod-ML (forecasts for 12 horizons, eq. 22), factor sorts, 1/N, minimum variance, market; “*” marks an extra tuning layer (Section 5.3)

Appendix: Case Study, Complete Results

- Table 2, Portfolio-ML: net Sharpe ratio 1.33, utility 0.086 (net return 15%, volatility 11%, cost 0.6% a year, turnover 0.25)
- Table 2, Multiperiod-ML*: net Sharpe 0.83, utility 0.020; Portfolio-ML needs no second tuning layer
- Table 3: posterior probability that Portfolio-ML has the higher utility: 99.9% against Static-ML* and Multiperiod-ML*, 100% against Markowitz-ML
- Figure 1(B) (Section 6.2): with \$10 billion and $\gamma = 10$, net excess return 14.6% at 11% volatility; with \$1 billion, 16.3% at the same volatility: size is a cost
- Figure 5: net of costs, quality is the most important theme and short-term reversal almost irrelevant; monthly autocorrelation -0.04 for 1-month reversal, 0.94 for book-to-market (Section 6.3): persistent signals are cheap to trade
- Limits stated by the authors: fees and other frictions, computing power and ML methods not available in real time early in the sample (Section 6.1); costs and risk treated as constant over the horizon of each optimisation (Section 5.1.3); among the smallest 20% of stocks Portfolio-ML ranks behind Multiperiod-ML* and Static-ML* (Figure 8)

Appendix: Case Study, the Course Replication

- Course data: 14 US stocks and 9 sector ETFs; five signals (1-month return, momentum from $t - 6$ and from $t - 12$ to $t - 1$, 1-year volatility, 6-month dollar volume); same cost rule, γ , wealth and grid; Σ with the same weights on the 23 assets; validation from 2011; out-of-sample returns February 2014 – August 2026 (151 months)
- Portfolio-ML: net Sharpe 0.57, utility 0.016, cost 0.6% a year, leverage 0.32; 1/N: net Sharpe 0.61, utility -0.26 (net volatility 30%)
- Utility: Static-ML -0.13 ; minimum variance -0.72 (cost 43% a year); Markowitz-ML extremely negative (leverage 18)
- Posterior probability of a higher utility for Portfolio-ML: 99.5% against Static-ML, 99.9% against minimum variance, 93.3% against 1/N
- Same ranking of cost-aware against cost-agnostic rules as Table 2; Portfolio-ML wins with small, slowly traded positions, not with a higher gross Sharpe ratio (0.68 against 0.87 for 1/N)
- Limits: 23 assets instead of 184–805, five signals instead of 115, no factor model for Σ , whose window reaches 10 years only in August 2014 (common prices from August 2004); return forecasts with $p \geq 2^6$ and three validation years (paper: 2^1 – 2^9 , ten years), no starred rivals
- Open extension: implementable frontiers for wealth of \$1, 10 and 100 billion and $\gamma \in \{1, 5, 10, 20, 100\}$, with 1/N's trading costs computed at each wealth level (Seminar 4, C3)

Appendix: The Discovery Loop, Reusable Prompts (1/2)

- Tools are interchangeable: an AI assistant (Claude, ChatGPT, Gemini, GitHub Copilot) and literature engines (Semantic Scholar, Elicit, Google Scholar)
- **Literature:** *"List peer-reviewed studies since 2015 comparing shrinkage or risk-based rules with $1/N$ out of sample; for each give DOI, N , T , cost assumption and test."* Resolve every DOI on Crossref and open the table cited
- **Hypotheses:** *"Give three competing explanations for a Sharpe gain over $1/N$ (better Σ , a low-volatility premium, luck) and the testable implication of each for turnover, subperiods and factor loadings."*
- **Data and code:** *"Write Python that reads the course's daily price files, aligns prices on common dates and runs a 60-month rolling GMV-LW and $1/N$; weights at t may use data up to $t - 1$ only."* Replicate first: $1/N$ 0.76, GMV-LW 0.20 on the combined universe

Appendix: The Discovery Loop, Reusable Prompts (2/2)

- **Robustness:** *"Draft a pre-registration: the family of tests (rules $\times$ universes $\times$ windows $\times$ costs), the error rate (FWER or FDR), the subperiods and a placebo, fixed before any result is seen."*
- **Critique:** *"Act as a referee: give the three strongest reasons my Sharpe gain is spurious and the test that would expose each."*
- **Report:** every specification tried, adjusted p -values, the power of the design, code and seeds; any specification chosen after seeing results is exploratory
- Portfolios aggregate **simple** returns and Sharpe ratios use **excess** returns: recompute one month step by step

Appendix: Further Study, Code and Data

Advanced courses on quantinar.com

- ☐  Hierarchical Clustering
- ☐  Minimum Spanning Tree
- ☐  Multivariate Statistical Analysis

Code and data

- ☐ All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_04
- ☐ Risk-free rate: [Kenneth French Data Library](#)

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